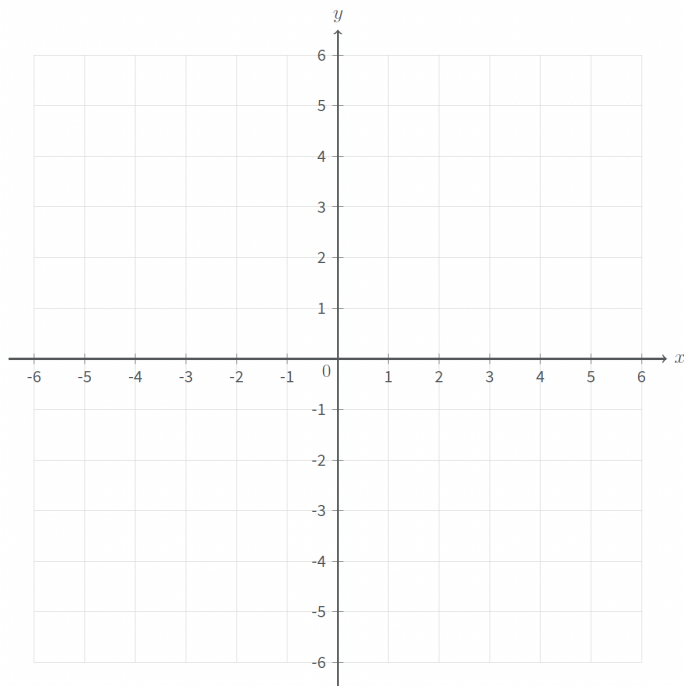
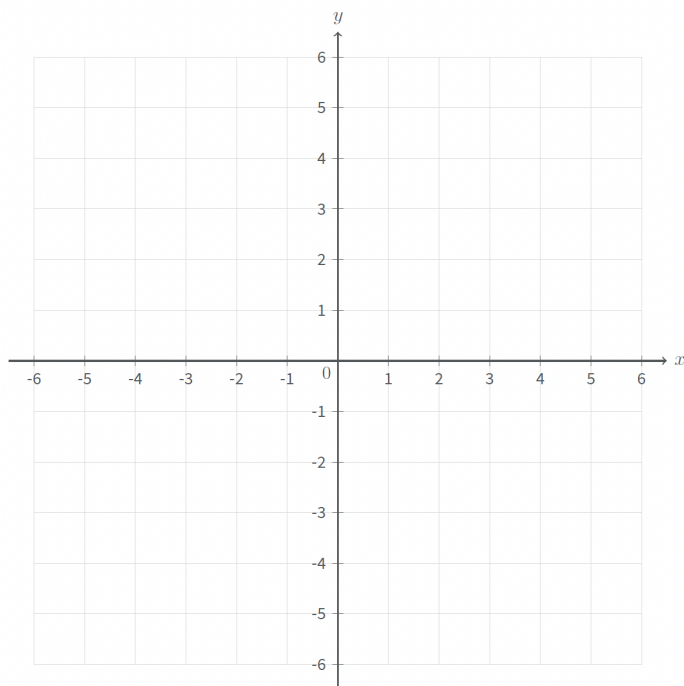


Day 11: Exponential Functions

1. Graph $y = 3^x$. Label any x and y -intercepts.



2. Graph $y = \left(\frac{1}{2}\right)^x$. Label any x and y -intercepts.



3. How would the graph of $y = (\frac{1}{2})^x$ and $y = (\frac{1}{2})^x + 3$ differ or be the same?
Draw a rough sketch of the two functions.

4. How would the graph of $y = (\frac{1}{2})^x$ and $y = (\frac{1}{2})^x - 2$ differ or be the same?
Draw a rough sketch of the two functions.

5. At what point, if any, does the graph of $y = 3^{2x-1}$ intersect the line $y = 27$?

6. Without graphing, determine if the two functions $f(x) = 4^{x+3}$ and $g(x) = 16^x$ intersect. If they intersect, write the point of intersection in the form (x, y) .

7. Uriel graphed the two functions $y = 3^{2x+2}$ and $y = 9 \cdot 9^x$ on Desmos and got the graph below. Why? What do you think is going on?

